

Blood Bank Inventory & Emergency Donor Matcher

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible eligible donors within a 10 km radius.	High	Pass: When an emergency request is flagged, the system identifies compatible eligible donors within 10 km and sends an SMS alert to each matching donor within 30 seconds. Fail: Any donor who does not match the required blood group/component or eligibility criteria receives the emergency alert.	Enables rapid response to critical blood shortages while limiting alerts to appropriate donors.
FR-002	Functional	The system shall allow the Blood Bank Manager to record, update, and view blood component inventory by blood group, component type, quantity, collection date, and expiry date.	High	Pass: A manager can create or update an inventory record, and the updated blood group, component type, quantity, collection date, and expiry date are displayed correctly when the record is retrieved.	Provides accurate, current visibility of available blood components.
FR-003	Functional	The system shall automatically identify and flag blood units that are expired or within the configured expiry-warning period.	High	Pass: A unit at or beyond its expiry date is flagged as expired, and a unit within the configured warning period is flagged for review. Fail: An expired unit remains available for emergency allocation.	Reduces the risk of allocating unsuitable or expired blood components.
FR-004	Functional	The system shall allow an authorized Emergency Requester to create an emergency blood request specifying the required blood group, component type, quantity, urgency, and request location.	High	Pass: A request cannot be submitted until all required fields are provided, and a successfully submitted request receives a unique request ID and emergency status.	Captures the information required to coordinate an emergency response.
FR-005	Functional	The system shall allow the Blood Bank Manager to allocate available blood units to a valid emergency request	High	Pass: After allocation, the selected units are associated with the emergency request and the	Ensures emergency requests are fulfilled using accurately tracked stock.

		and update the remaining inventory quantity.		available inventory quantity decreases by exactly the allocated quantity. Fail: The system permits allocation when insufficient compatible inventory remains.	
NFR-001	Nonfunctional — Performance & Security	The inventory ledger shall maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.	High	Pass: Under simulated concurrent allocation tests, each blood bag can be allocated to at most one request, no duplicate allocation records are created, and all committed inventory transactions preserve the correct final quantity.	Prevents double allocation and protects the integrity of critical blood inventory.
NFR-002	Nonfunctional — Performance	The system shall process emergency matching and notification requests with predictable performance under peak operational load.	High	Pass: In a peak-load test representing 500 concurrent emergency-matching requests, at least 95% of valid matching operations produce the donor-notification result within 30 seconds, without failed or duplicated notifications.	Supports timely emergency response when demand is highest.

Use Case Flow

Main Success Scenario

1. The Emergency Requester enters the required blood group, component type, quantity, urgency, and request location.
2. The system validates the emergency request and assigns a unique request ID.
3. The system cross-references the request against real-time blood bank inventory.
4. The system identifies compatible, non-expired blood units that satisfy the requested component and quantity.
5. The Blood Bank Manager reviews and confirms the proposed allocation.
6. The system atomically allocates the selected blood units and updates the remaining inventory quantity.
7. If the available inventory does not fully satisfy the emergency request, the system identifies the remaining shortage.
8. The system identifies eligible donors whose blood group/component compatibility and location satisfy the emergency matching rules.
9. The system broadcasts an SMS emergency alert to compatible eligible donors within a 10 km radius.
10. The system records the allocation and notification events and displays the emergency request status to the Emergency Requester.

Alternate Flow: Insufficient Compatible Inventory

1. At Step 4, the system determines that insufficient compatible, non-expired inventory is available.
2. The system records the available quantity and calculates the remaining shortage.
3. The system allocates only the confirmed available quantity without allowing the inventory quantity to become negative.
4. The system triggers the donor-matching process for the outstanding shortage and sends SMS alerts to compatible eligible donors within the 10 km radius.
5. The Emergency Requester is shown the partial-allocation status and remaining shortage.

Key Acceptance Checks

- Unique Request ID: An emergency request receives a unique request ID after successful submission.
- Expired Units Excluded: Expired blood units are excluded from emergency allocation.
- Donor Notification: Compatible eligible donors within 10 km are notified within 30 seconds of the emergency flag.
- No Double Allocation: A blood bag cannot be allocated to two requests during concurrent allocation attempts.

UML Use Case Diagram

Actors

- Emergency Requester
- Blood Bank Manager
- SMS Gateway

Use Cases

- UC-01: Create Emergency Blood Request
- UC-02: Check Blood Bank Inventory
- UC-03: Allocate Blood Units
- UC-04: Match Eligible Donors
- UC-05: Send Emergency SMS Alerts

Associations

- Emergency Requester → UC-01: Create Emergency Blood Request
- Emergency Requester → UC-04: Match Eligible Donors
- Blood Bank Manager → UC-02: Check Blood Bank Inventory
- Blood Bank Manager → UC-03: Allocate Blood Units
- SMS Gateway → UC-05: Send Emergency SMS Alerts

Stereotypes

- UC-01 Create Emergency Blood Request «include» UC-02 Check Blood Bank Inventory
- UC-03 Allocate Blood Units «include» UC-02 Check Blood Bank Inventory
- UC-04 Match Eligible Donors «include» UC-05 Send Emergency SMS Alerts
- UC-05 Send Emergency SMS Alerts «extend» UC-04 Match Eligible Donors



